

Product Requirements Document (PRD) - Multi-Agent Translation Orchestration

1. Product Vision

Develop an enterprise-grade autonomous system that guarantees the conceptual integrity and strategic intent of high-stakes managerial manifestos as they traverse multi-lingual pipelines, utilizing advanced AI governance and orchestration principles.

2. Functional Requirements

- **Level 4 Autonomous Orchestration:** A central engine managing a multi-node chain (EN-FR-HE-EN) with automated state tracking.
- **Pre-emptive Accuracy Control (Glossary Injection):** A RAG/Tool-use mechanism to inject hardcoded lexicons before translation to lock in critical philosophical tokens.
- **Human Trust Viability Score (HTVS):** A specialized metric to evaluate the preservation of "Technical Flavor" and rhetorical intent, with a mandatory qualitative threshold of 95%.
- **Active Feedback Loop:** Autonomous detection of "Semantic Smoothing" with dynamic prompt injection for iterative self-correction.
- **Managerial Audit Trail:** Detailed logging of vector-distance checks and inter-node analytical justifications.

3. Non-Functional Requirements

- **Multi-Persona Governance Ecosystem:** Integration of the Executive Review Board (ERB), Prompt Engineering Analyst (PEA), and AI Governance Officer (AIGO).
- **Context Integrity:** Utilization of Short-Term Context Memory to manage the context window across long interaction chains.
- **Lost-in-the-Middle Mitigation:** Strategic prompt structuring to ensure high visibility of constraints.
- **Human-in-the-Loop (HITL) Validation:** Final supervisory layer for strategic alignment before organizational dissemination.

4. Success Metrics (Evaluated KPIs)

- **Conceptual Integrity Score:** Evaluated at ~99% for target keywords (e.g., 'Orchestrate').
- **HTVS Certification:** Minimum qualitative assessment of 95% for executive distribution.
- **Semantic Drift (Vector Distance):** Estimated cosine similarity maintained at >0.95 for corrective iterations.
- **Governance Compliance:** Estimated ~98% compliance with organizational brand and rhetorical standards.